

2025 Calculus Mid Exam

《Chapter Review Infinite Series》

- Notice:**
1. Make sure that you have filled the form on the left side of seal line.
 2. Write your answers on the exam paper.
 3. This is a close-book exam.
 4. The exam with full score of 100 points lasts 120 minutes.

Question No.												Sum
Score												

1. The series absolute convergence or conditional convergence (20 points):

(1) $\sum_{n=1}^{\infty} (-1)^n \frac{1}{na^n} \quad (a \geq 1);$ (2) $\sum_{n=1}^{\infty} (-1)^n a^n \sin \frac{1}{2^n} \quad (a \geq 0);$

(3) $\sum_{n=2}^{\infty} \frac{(-1)^n}{(-1)^n + n} ;$ (4) $\sum_{n=2}^{\infty} (-1)^n \tan(\sqrt{n^2 + 2} \pi) ;$

2. Find the sum of power series (20 points):

(1) $\sum_{n=1}^{\infty} \frac{x^n}{n+1} \quad S(x) ;$ (2) $\sum_{n=0}^{\infty} \frac{1}{n!} x^{3n+4} \quad S(x) ;$

(3) $\sum_{n=1}^{\infty} \frac{2n-1}{3^n} \quad S(x) ;$ (4) $\sum_{n=2}^{\infty} \frac{1}{n^2-1} x^n \quad S(x)$

3. Find the Taylor series for the function (10 points):

(1) $f(x) = \frac{1}{x^2 - 3x + 2}, \quad \text{at } x = 5;$

(2) $f(x) = \frac{1}{4} \ln \frac{1+x}{1-x}, \quad \text{at } x = 0$

4. Find the radius and convergence set of power series (10 points):

(1) $\sum_{n=1}^{\infty} \frac{1}{n[3^n + (-2)^n]} x^n$

(2) $\sum_{n=2}^{\infty} \frac{n^2 + 1}{3^n n!} x^n$